

Activity 11 Pre-Lab — Synthesis of Luminol (Forensic Chemistry)

[Your name] & [partner]
[MM/DD/YY]

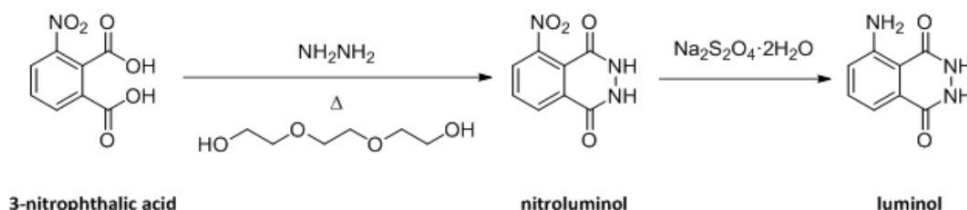
Organic Lab · two-step synthesis + chemiluminescence

■ Black = write this in your notebook ■ Blue = context, don't copy it Copy the header onto every page.

Purpose

Synthesize **luminol** from 3-nitrophthalic acid in two steps (hydrazide formation, then nitro → amine reduction), then test its **chemiluminescence** with an iron “blood mimic” + hydrogen peroxide, comparing room-temp vs chilled.

🔥 DRAW — the 2-step synthesis (from the procedure)

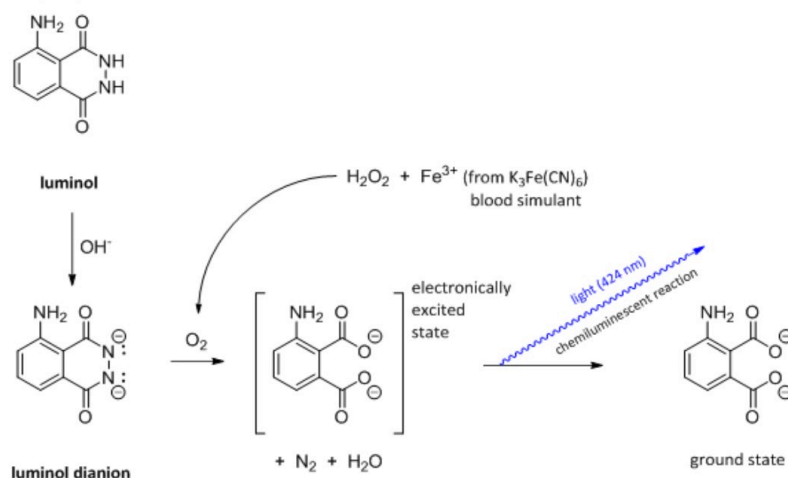


Step 1: 3-nitrophthalic acid

+ hydrazine → 3-nitrophthalhydrazide (triethylene glycol, ~200 °C). Step 2: Na₂S₂O₄ reduces –NO₂ → –NH₂ → luminol; acetic acid precipitates it.

🔥 DRAW — the chemiluminescence + its mechanism (required)

Figure 2.1: Physical Chemiluminescence of Luminol



Directions require the mechanism for

the boxed step — leave room. luminol → dianion (OH[−]) → oxidized by H₂O₂/Fe³⁺ to an excited 3-aminophthalate → relaxes, emitting blue light.

Chemical Reagents & Safety Table

CONTEXT ▶ Required (8). Typical SDS values — cross-check the SDS tile. **Hydrazine is toxic & a carcinogen.**

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
3-Nitrophthalic acid	Starting material	211.13	~1 g	Off-white / pale-yellow solid	irritant
Hydrazine (8% aq)	Forms the hydrazide ring	32.05	2 mL	Colorless aqueous liquid	toxic corrosive health environ.
Triethylene glycol	High-boiling solvent	150.17	3 mL	Colorless viscous liquid; bp 285 °C	— low hazard
Sodium hydrosulfite dihydrate	Reduces –NO ₂ → –NH ₂	174.11	3 g	White powder	flammable irritant
Sodium hydroxide (3 M aq)	Base: reduction + dianion	40.00	5 mL (+2 mL Part 2)	Colorless aqueous solution	corrosive

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
Glacial acetic acid	Acidify → precipitate luminol	60.05	2 mL	Colorless liquid; bp 118 °C	  flammable corrosive
Potassium ferricyanide (3% aq)	Catalyst — blood/iron mimic	329.24	4 mL	Red crystalline solid / soln	— low hazard (no acid!)
Hydrogen peroxide (3%)	Oxidant for the glow	34.01	4 mL	Colorless liquid	 irritant

GRAB FIRST: 50-mL RBF + clamp · heating mantle + **Variac** + raised lab jack · simple-distillation head + **high-temp thermometer ($\geq 200^\circ\text{C}$)** + condenser + receiver · boiling stones · Erlenmeyer + steam bath (heat 15 mL water) · ice bath · Büchner + filter flask + vacuum · 3-mL syringe + 25-mL Erlenmeyer (hydrazine) · stir bar + stir plate · spatula · **Part 2:** 125-mL Erlenmeyer + 2× 50-mL beakers.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Reaction 1 — make nitroluminol

- First:** heat **15 mL water** (+ boiling stones) in an Erlenmeyer on a steam bath — you need it hot at step 8.
- Put **~1 g 3-nitrophthalic acid** in a **50-mL RBF (no stir bar yet)**. Clamp it; lower into a heating mantle on a raised lab jack so the flask **touches the mantle ceramic**.
- Nitrile gloves: take a **3-mL syringe** in a 25-mL Erlenmeyer to the hydrazine station; draw **2 mL 8% hydrazine**; carry back and transfer it into the flask.

Hydrazine is corrosive & a carcinogen. No skin/eye/clothing contact. Rinse the syringe 2× w/ DI water into the waste beaker → syringe to solid waste → **fresh gloves**.

- Add a few **boiling stones**; heat gently at **Variac 75–80%** until the solid acid dissolves.
- Add **3 mL triethylene glycol**.
- Build a **simple distillation**. The **high-temp thermometer must be submerged in the solution but NOT touching the flask bottom**. **Don't adjust it while hot — pull it out, tweak, replace. If it touches bottom it breaks as temp rises.**
- Instructor inspects → **Variac 80%** → heat until the thermometer reads **200 °C** = reaction complete. **Water distilling out is discarded; boiling it off drives the ring formation.**
- Remove from heat; **cool to ~100 °C** (crystals often appear); add the **15 mL hot water** from step 1.
- Cool the flask in an **ice bath** to near room temp → **vacuum filter** the light-yellow nitro compound. **No need to dry it before Reaction 2.**

Reaction 2 — reduce to luminol

- Transfer the crude nitroluminol **back into the same (uncleaned) flask**. Clamp it and **add a stir bar**.
- Stirring, add **5 mL 3 M NaOH** → deep **brown-red** solution. Then add **3 g sodium hydrosulfite dihydrate** ($\text{Na}_2\text{S}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$). **Break up chunks with a spatula; wash any solid down the walls with a little water from a pipette.**
- Slide the **heating mantle** back under the flask (lower the jack to fit it). Stirring, heat to **boiling at Variac 60–65%**; once boiling, stir **5 min**. **Some reduction product may separate out — that's fine.**
- Variac off**, lower the jack, remove the mantle.
- Add **2 mL glacial acetic acid**; **cool on an ice bath** with stirring. **Vacuum filter** the light-yellow **luminol**. **Keep it moist — you use it wet in Part 2.**

Part 2 — Chemiluminescence **make all three solutions first, then set aside**

- Solution 1 (blood mimic)** — 125-mL Erlenmeyer: **4 mL 3% $\text{K}_3\text{Fe}(\text{CN})_6$ + 32 mL water**.
- Solution 2 (luminol dianion)** — 50-mL beaker: your **moist luminol** (~40–60 mg dry) + **2 mL 3 M NaOH + 18 mL water**; swirl gently to dissolve.
- Solution 3 (peroxide)** — 50-mL beaker: **4 mL 3% H_2O_2** .
- Pair with another group: **Group A = room temperature; Group B = chill all three solutions** in ice baths several minutes.
- At the same time as your partner group, simultaneously pour Solution 2 AND Solution 3 into Solution 1.** Record **intensity and duration** of the glow; compare RT vs chilled.

Experiment Notes (in lab)

measurement	value
Mass 3-nitrophthalic acid	g
Rxn 1 max temp reached	°C
Mass luminol (or moist est.)	g
% yield	

Rxn 1 — color changes / when crystals appeared:

Nitro compound appearance:

Rxn 2 — brown-red? on adding acetic acid?

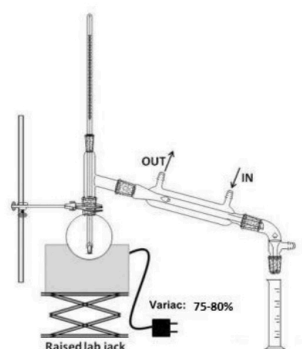
Glow test

condition	color	brightness	duration
Room temp (A)			
Chilled (B)			

WRITE

theoretical luminol from 1 g 3-nitrophthalic acid ($\div 211.13 \rightarrow \times 177.16$) $\rightarrow$ % yield.

Setup — draw this



CONTEXT (don't copy)

Why 200 °C? the hydrazide ring only closes at high temp — triethylene glycol (bp 285 °C) lets you get there, and boiling off the water pushes the equilibrium to product.

Why the glow? H_2O_2 (with Fe^{3+} catalysing) oxidizes the luminol dianion to an excited 3-aminophthalate; dropping to ground state emits blue light. Colder = slower reaction = dimmer but longer-lasting glow.

Never mix ferricyanide with acid — releases toxic gas.

Rxn 1: simple distillation (thermometer submerged, ≥ 200 °C)

Clean-up & Conclusions

- Hydrazine rinses $\rightarrow$ **labeled hydrazine waste**; distillate water & filtrates $\rightarrow$ aqueous waste; keep ferricyanide away from acid; return glassware; **locker check-out** + group clean-up (empty drawer, new liner, drying tube emptied & rinsed).
- Conclusions** (~4 sentences [after lab](#)): overall % yield; what each step did (ring formation, then $-\text{NO}_2 \rightarrow -\text{NH}_2$); what the glow test showed and how temperature changed intensity/duration; a loss/error source.